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Comprehensive understanding of SiO_2-promoted Fe Fischer-Tropsch synthesis catalysts: Fe-SiO_2 interaction and beyond

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ABSTRACT

The physico-chemical properties of SiO_2-promoted Fe-based Fischer-Tropsch synthesis catalysts were traditionally considered to be governed by Fe-SiO_2 interaction. Here we found that the configuration between SiO_2 and iron species also played a pivotal role in determining the structure and thus FTS performance of the catalysts. Fe@Si catalyst with embedded structure, impregnated Fe/Si catalyst as well as precipitated Fe-Si catalyst was fabricated by different methods respectively, and they were thoroughly characterized by multiple techniques. The results indicated that, despite the weaker Fe-SiO_2 interaction, the construction of SiO_2 shell outside the iron species core in Fe@Si catalyst strongly inhibited the reduction of iron oxides due to the confining effect of SiO_2 shell, which increased the diffusion resistance of H_2O generated in the reduction process. However, the sintering of iron species was effectively hindered even during FTS reaction by the physical separation of SiO_2 shell. In contrast, iron species in impregnated Fe/Si and precipitated Fe-Si catalyst experienced aggregation with different degrees. The reduction behavior of these two catalysts were well correlated with the strength of Fe-SiO_2 interaction. The FTS performance showed that both Fe@Si and Fe/Si catalyst exhibited higher initial activity but deactivated gradually, while the activity of Fe-Si catalyst displayed an opposite trend. These phenomena were discussed in terms of the variation of iron carbides content at different stages, which was determined not only by Fe-SiO_2 interaction, but also by the manner that SiO_2 and iron species constructed. The present study contributed a new understanding of the structure-performance relationship for SiO_2-promoted Fe-based FTS catalysts.

1. Introduction

Fischer-Tropsch synthesis (FTS) is a vital technology in heterogeneous catalysis, as it is widely used for the production of clean fuels and high value-added chemicals from syngas (H_2/CO), which is readily derived from coal, biomass, or natural gas. Highly efficient catalysts are considered as the key to enhance the process-economic efficiency (higher conductivity and higher energy efficiency) of FTS technology [1,2]. Fe-based catalysts have been proven to be more favorable than other catalyst system due to the tunable product distribution, lower cost, and higher water-gas shift (WGS) activity [3–5].

Promoters are usually employed for commercial Fe-based catalysts since pure Fe is not suitable for industrial application due to its poor activity, undesired selectivity, and low mechanical strength [6–8]. For example, Cu has been identified as a key promoter for improving the reduction and carburization ability of Fe and thus increasing the number of active sites [9]. Alkali promoters like K are introduced to hinder the formation of gaseous products, shifting the products slate to the higher carbon hydrocarbons [10]. The incorporation of structural promoters or supports is considered an efficient way to maintain the integrity of the catalysts structure [11,12]. SiO_2 is the most frequently studied promoter and to some extent, the satisfied ones for Fe-based FTS catalysts [13–15]. Promotion of SiO_2 is initially designed to enhance the mechanical strength of catalyst, while exerting diverse influence on the physico-chemical properties and performance [16–18]. Specifically, it was reported that the presence of SiO_2 leads to an increase in the dispersion state and attrition resistance [19], as well as a decrease in reduction and carburization extent [6]. Moreover, the lower activity and

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improved stability is usually observed for SiO_2-containing catalysts [20]. Historically, these phenomena were attributed to the ambiguous term, Fe-SiO_2 interaction. Our previous study indicated that the so-called Fe-SiO_2 interaction could be understood in terms of the Fe-O-Si structure formation, which originates from the dehydration of surface OH located between Fe species and Si during thermal treatment [21]. This unique Fe-O-Si structure hinders particle agglomeration during the calcination process and accounts for the high dispersion state of the iron oxides. However, the strengthened Fe-O bond also makes the O removal more difficult during reduction and carburization, which lowers the reduction and carburization degree and ultimately degrades the activity.

In the aforementioned results, the iron-silica composites were mainly fabricated via the impregnation or precipitation method, where no specific configuration between Fe and SiO_2 species was formed due to their random distribution. Despite the presence of strong Fe-SiO_2 interaction, the primary drawback of traditional Fe/SiO_2 catalysts lies in the sintering of the Fe species during FTS reaction. Metal@SiO_2 catalysts, a unique catalyst system where the active metal located inside the SiO_2 shell which inhibit the agglomeration of the active phase, have gained increasing attention in recent years [22,23], and also found application in Co-based FTS catalysts. Xie et al. [24] found that the SiO_2 shell could prevent the Co particles from sintering and guaranteed their high dispersion state during the FTS reaction, compared to the impregnated Co/SiO_2 catalyst, which was responsible for the stable CO conversion of Co@m-SiO_2 catalyst. It is reasonably expected that the Fe particle sintering, a common phenomenon in Fe-SiO_2 FTS catalysts prepared by impregnation or precipitation method, would be alleviated due to the confining effect of SiO_2 shell in Fe@SiO_2 catalyst. However, the related research was scarcely reported yet. One thing to be noted is that the formation of Fe-SiO_2 interaction during the construction of SiO_2 shell is inevitable, which would exert extensive influence on the dispersion state and reduction behavior of the Fe species confined. Therefore, the attempt to clarify the effect of Fe-SiO_2 interaction and the confining effect of SiO_2 shell on the physico-chemical properties and FTS performance of core-shell Fe-SiO_2 catalyst is necessary.

Herein, we aimed to examine the effect of Fe-SiO_2 interaction as well as SiO_2 shell confinement on the physico-chemical properties and catalytic performance of SiO_2-promoted Fe-based FTS catalysts. Three catalysts with the same Fe/Si ratio were prepared using different protocols, 1) The Fe@Si catalyst with an embedded structure was obtained from the Stöber process, 2) The Fe/Si catalyst was derived from the impregnation method, and 3) The Fe-Si catalyst was prepared via the co-precipitation technique. The catalysts were studied and compared by employing a combination of multiple tools, such as the N_2 physisorption, Fourier-transform infrared spectroscopy (FT-IR), Mössbauer effect spectroscopy (MES), transmission electron microscopy (TEM), and H_2 temperature-programmed reduction (H_2-TPR). Then, their performance was subsequently tested in an FTS reaction. This study was intended to provide in-depth insight and comprehensive understanding of the structure-performance relationship of SiO_2-promoted Fe-based FTS catalysts and subsequently offer new information for innovative catalyst designs.

2. Experimental

2.1. Catalyst preparation

Fe_2O_3 nanoparticles were prepared according to the reported procedure with some modifications [25]. In a typical synthesis, 7.39 g of Fe $(NO_3)_3 \cdot 9H_2O$ and 7.21 g of PVP M.W. = 58,000 were dissolved in 800 mL of anhydrous ethanol at room temperature. After stirring for 5 h, the yellow suspension was transferred into a stirrer stainless autoclave with a total volume of 1000 mL. The reaction mixture was heated to 180 °C under stirring with a constant heating rate of 1 °C/min and kept for 18 h. Finally, the resulting solution was cooled to room temperature and slowly poured into a beaker for the next step.

The embedded catalyst was synthesized by the Stöber method with some modifications [26]. In a typical synthesis, 144 mL of $NH_3 \cdot H_2O$ (25%) was added in a solution containing 1656 mL of deionized water and 1276 mL of EtOH under continuously stirring. Secondly, 6 g of cetyltrimethylammonium bromide (CTAB) as template was added to form the mesoporous SiO_2 shell. After stirring for 5 h, 25 mL of tetraethoxysilane (TEOS) was added into the above mixture during 6 h, and then strongly stirred for 48 h at room temperature. The solid products were collected by filtration and washed with distilled water three times, and then dried at 120 °C for 24 h in air. The template was removed by calcination at 550 °C for 6 h in air. The as-prepared catalyst was named as Fe@Si. The synthesis and post-treatment procedure of m-SiO_2 support was similar to that of Fe@Si without adding Fe_2O_3 NPs. The obtained support was named as m-SiO_2.

The impregnated catalyst was prepared by incipient wetness impregnation (IWI) method [27]. Typically, the 10.0 g of as-prepared m-SiO_2 was impregnated with the ethanol solution of $Fe(NO_3)_3$ (2 M) for 5 h and then the mixture was transferred to the Teflon-lined stainless steel autoclave and heated to 180 °C for 18 h. The final product was filtered and washed with distilled water and anhydrous ethanol. Finally, the sample was dried and calcined at 550 °C for 6 h in air flow. The as-prepared catalyst was named as Fe/Si.

The precipitated catalyst was obtained by the co-precipitation technique [28]. In a typical synthesis, 20.6 g of the $Fe(NO_3)_3 \cdot 9H_2O$ and 31.7 g of silica sol (SiO_2, 30 wt.%) were dissolved in distilled water; and 200 mL of $NH_3 \cdot H_2O$ (25 wt.%) was dissolved in 500 mL of distilled water. The two solutions were added simultaneously into the beaker to maintain the pH at a constant value of 9.0 ± 0.1 at 85 °C. The resulting solid was recovered by filtration, washed extensively with distilled water, dried at 120 °C, and calcined at 550 °C for 6 h. The as-prepared catalyst was named as Fe-Si.

2.2. Catalysts characterization

The bulk chemical composition of the catalysts was determined by inductively coupled plasma-atomic emission spectrometer (ICP-AES).

The textural properties were measured by N_2 physisorption at −196 °C, using ASAP 2420 instrument (Micromeritics, USA). Each catalyst was degassed under vacuum at 90 °C for 120 min and 350 °C for 480 min prior to the measurement, and the surface areas were calculated using the BET method.

The FT-IR spectra were recorded with a VERTEX 70 (Bruker, Germany) FT-IR spectrophotometer. Samples were mixed with KBr at the ratio of 1/50 and pressed into translucent disks at room temperature. All spectra were taken in the range of 4000−400 cm^{-1} at a resolution of 4 cm^{-1}.

Powder X-ray diffraction (XRD) patterns of different catalysts were determined on a Bruker D8 ADVANCE diffractometer using Co Kα radiation (λ =1.79026 Å, 35 kV and 40 mA) in the scan range of 30° to 90° with a step size of 0.04°. The phase identification was carried out by comparing the diffraction peaks with the standard JCPDF.

MES experiments were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany) driven with a triangular reference signal at 50 K. Data analysis was performed using a nonlinear least-squares fitting routine that modeled the spectra as a combination of singlets, quadruple doublets, and magnetic sextets based on a Lorentzian lineshape profile. The components were identified based on their isomer shift (IS), quadruple splitting (QS), and magnetic hyperfine fields (Hhf). Magnetic hyperfine fields were calibrated with the 330 kÖe field of α-Fe foil.

TEM images of the fresh and spent catalysts were recorded by using an FEI Talos F200X, operating at the acceleration voltage of 200 kV. Prior to analysis, the samples were dispersed in ethanol for 20 min, and then deposited onto carbon membrane for observation. The HADDF-STEM images were acquired in the scanning transmission electron microscopy (STEM) mode.

The hydrogen temperature-programmed reduction (H_2-TPR) experiment was performed with an Autochem II 2920 equipment (Micromeritics, USA). In a typical process, the catalyst (about 100 mg) was loaded into a U-type tube reactor and ramped from 50 to 800 °C at 10 °C/min under a flow of 10% H_2/90% Ar (50 mL/min). The variation of gas concentration was recorded by a thermal conductivity detector (TCD). The generated water during the reduction treatment was removed by isopropyl alcohol gel (−97 °C).

2.3. Catalyst testing

Before the catalytic test, each catalyst was activated at 350 °C for 24 h in a flow of syngas (H_2/CO = 2) at atmospheric pressure. After the activation step, the bed was cooled to 200 °C, and then the syngas flow (H_2/CO = 2) was gradually introduced to the reaction bed. The catalytic performance of different catalysts for CO hydrogenation was tested in a fixed-bed reactor operating at 2.0 MPa, 280 °C, H_2/CO = 2 and GHSV =5000 $mL\cdot h^{-1}\cdot g_{cat}^{-1}$. The effluent gas from the reactor was analyzed by on-line gas chromatography (GC, 7890B, Agilent). CO conversion, CO_2 and hydrocarbon selectivity were calculated after 64 h on stream and defined in previous papers [5,29,30]. The carbon balance was higher than 95%.

3. Results and discussion

3.1. Compositions and morphology of the fresh catalysts

A two-step strategy involving solvothermal synthesis and the Stöber method was employed to fabricate the Fe@Si catalyst. In the first stage, polyvinylpyrrolidone (PVP) was briefly used as a stabilizer in order to eliminate aggregation of the Fe_2O_3 nanoparticles [31,32]. Next, the SiO_2 shell was formed by hydrolyzing tetraethoxysilane (TEOS) on the Fe_2O_3 core surface, after which the Fe@Si catalyst was successfully prepared. The reference catalysts Fe/Si and Fe-Si were prepared by the IWI and co-precipitation method, respectively, as described in the experimental section. The composition of the Fe@Si, Fe/Si, and Fe-Si catalysts are listed in Table 1. The morphologies of all the prepared catalysts were characterized by using TEM, and the corresponding elemental distribution was confirmed by energy-dispersive X-ray (EDX) mapping (Fig. 1). Results show that Fe_2O_3 particles are successfully (the average size is 50.4 nm) embedded into SiO_2 matrix for the Fe@Si catalyst. Compared with Fe/Si and Fe-Si catalysts, each nanoparticle in the embedded structure is isolated by a highly porous SiO_2 shell. The Fe@Si catalysts (Fig. 1a) exhibit good stability without evident sintering during the high temperature calcination process, due to protection from the SiO_2 shell. For Fe/Si catalyst as shown in Fig. 1b, however, serious agglomeration of Fe_2O_3 particles is observed. The result indicates that the primary defect of impregnation method is extensive agglomeration and sintering of Fe particles during the preparation process as well as activation and reaction process (see below). Also, one can find that the iron species distribute evenly in the SiO_2 matrix in Fe-Si catalyst (Fig. 1c), which is caused by the strong Fe-SiO_2 interaction formed in co-precipitation process as detailed later. The crystal phase compositions of catalysts are assessed by XRD. As shown in Fig. S1, the only detectable phase in the XRD patterns of the Fe@Si and Fe/Si catalysts is well-crystallized hematite (α-Fe_2O_3) with characteristic diffraction peaks at 2θ = 38.7°, 41.7°, 63.9°, and 74.1° [33]. By contrast, the Fe-Si catalyst displays broad diffraction peaks at 2θ = 42° and 75°, which are assigned to the amorphous iron oxides with small crystallites [34].

Table 1
Element content and textual properties of Fe@Si, Fe/Si and Fe-Si catalysts.

Catalysts	Fe content [wt.%]	BET surface areas [m^2/g]	Pore volume [cm^3/g]	Pore size [nm]
Fe@Si	30.9	652	0.64	2.62
Fe/Si	31.2	642	0.55	2.55
Fe-Si	30.2	295	0.46	6.28

3.2. Textural properties of the fresh catalysts

Fig. 2a shows the N_2 adsorption-desorption isotherms of the Fe@Si, Fe/Si, and Fe-Si catalysts. It is found that all the samples exhibit typical type IV isotherms [35], indicating that all the samples are mesoporous materials. Specifically, Fe@Si and Fe/Si exhibit H1-type hysteresis loops with capillary condensation steps at relative pressures (P/P0) between 0.4 and 0.9, which indicates the presence of uniform cylindrical pores in these two samples. The Fe-Si catalyst depicts H3-type hysteresis loops with the capillary condensation at a higher P/P0 (0.6-0.9), suggesting a relatively larger size mesopore existing there. The textural properties of catalysts are calculated from the isotherms, and summarized in Table 1. Apparently, the Fe@Si and Fe/Si catalysts exhibit higher BET surface area and smaller pore size than that of the Fe-Si catalyst. As mentioned in the experimental section, in the preparation of Fe@Si and Fe/Si catalysts, CTAB was used as the structure directing agents which lead to the formation of small and uniform pores [36], accounting for the relative higher BET surface area and smaller average pore size of these two catalysts. For Fe-Si catalyst, the larger and non-uniform pore (Fig. 2b) is mainly formed through the randomly dehydration process between the surface hydroxyl of FeOOH and SiO_2 during the thermal treatment, which is responsible for the lower BET surface area and larger average pore size.

3.3. Fe-SiO_2 interaction of the fresh catalysts

The Fe-SiO_2 interaction is characterized by FT-IR and the fitting results are summarized in Table S1. The spectra corresponding to the Fe@Si and Fe/Si catalysts show a broad peak around 538 cm^{-1}, which is characteristic of the Fe-O band in the α-Fe_2O_3 phase [37]. The peak centered at 1120 cm^{-1} is attributed to the asymmetric Si-O-Si stretching vibration [38,39]. Previous studies have reported that the presence of Fe-O-Si bonds is a key characteristic of the Fe-SiO_2 interaction in the catalysts, and the position as well as intensity of Fe-O-Si structure could be tentatively employed as a descriptor for the strength of Fe-SiO_2 interaction (lower wave number and lower intensity means weaker interaction) [21]. As shown in Fig. 3, all the catalysts provide the typical Fe-O-Si structure (adsorption band in the range of 957−977 cm^{-1}), which confirms the existence of Fe-SiO_2 interaction in the catalysts [40]. However, it should be pointed out that, due to low iron content herein, the Fe—O—Si adsorption peak exhibits a relatively low intensity compared with that in previous studies. Compared with the Fe/Si and Fe-Si catalysts, the Fe—O—Si stretching frequency of Fe@Si catalyst at ca. 957 cm^{-1} shows a red-shift. In addition, the Si—O—Si stretching vibration of the Fe/Si and Fe-Si catalysts are blue-shifted due to the large number of Fe species interacting with SiO_2 to form Fe—O—Si complexes. Alternatively, the curves from 850 to 1400 cm^{-1} have been fitted with two peaks (peak I represents the intensity of Fe—O—Si bond and peak II is Si-O-Si bond, respectively) in order to determine the relative intensity of Fe-SiO_2 interaction of each catalyst (the insert in Fig. 3). The results indicate that the relative intensity of Fe—O—Si structure exhibits the order of Fe@Si (I/II ratio = 0.077) < Fe/Si (0.109) < Fe-Si (0.191). One can also find that no obvious Fe—O bond stretching vibration bands are found in the Fe-Si catalyst, implying the high dispersion state of iron oxides resulted from the strong Fe-SiO_2 interaction. Based on the above analysis, it can be concluded that the order of Fe-SiO_2 interaction strength is as follows: Fe-Si > Fe/Si > Fe@Si.

The Fe-SiO_2 interaction strength for the present three catalysts are considered to be governed by the preparation method since their composition is almost the same. Previous studies concluded that the formation of Fe-SiO_2 interaction originated from the dehydration process between the hydroxyl groups on the surface of SiO_2 and iron oxyhydroxide [21,41]. Therefore, one can reasonably draw the conclusion

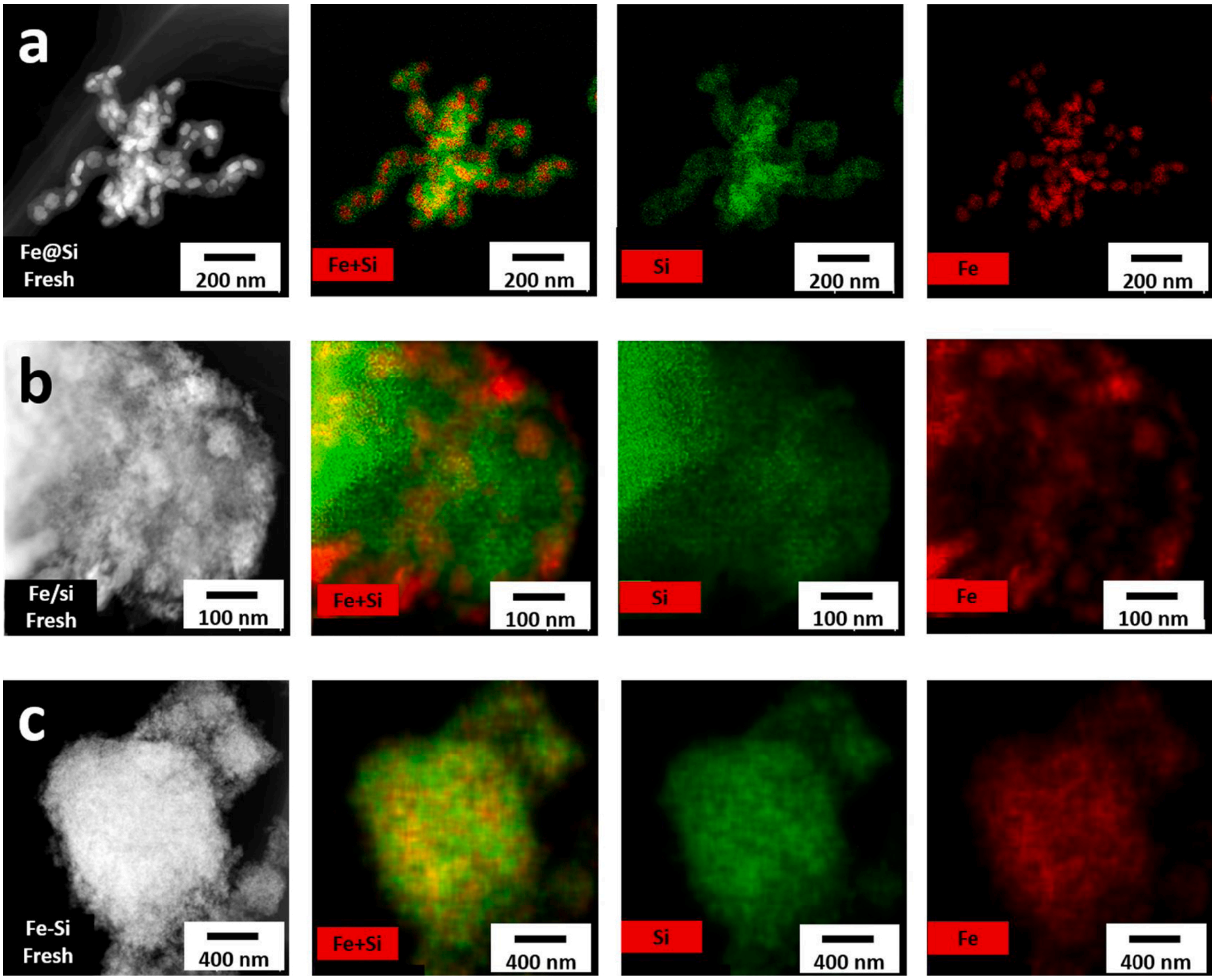

Fig. 1. TEM and high-angle annular dark-field STEM (HAADF-STEM) images of fresh catalysts; the elemental maps of the same particle for Si (green) and Fe (red), respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

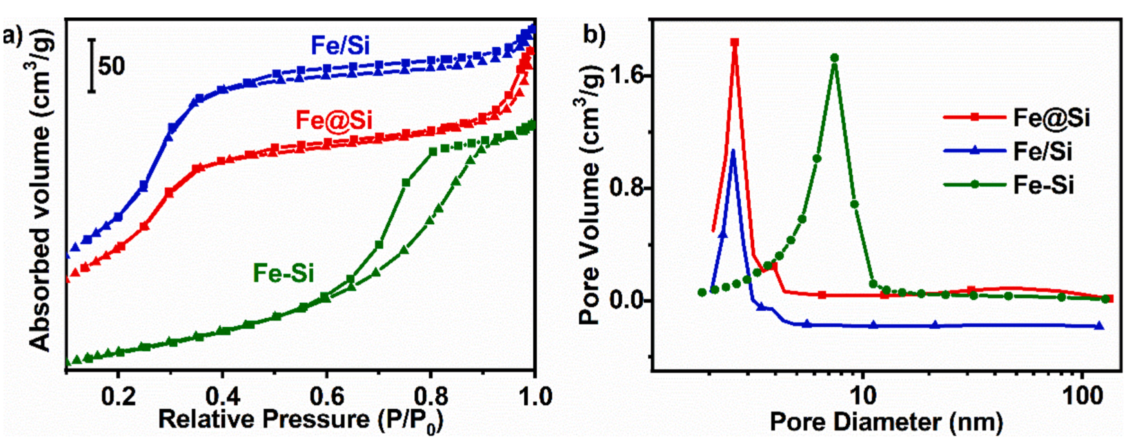

Fig. 2. N_2 desorption and adsorption isotherms of fresh Fe@Si, Fe/Si and Fe-Si catalysts (a) and pore-size distributions of Fe@Si, Fe/Si and Fe-Si obtained by the BJH method (b).

that the Fe-SiO_2 interaction strength is mainly determined by the quantity of hydroxy groups existed in the system for a given preparation method. For Fe-Si catalyst, there are plenty of hydroxyl groups on the surface of Silica sol as well as iron oxyhydroxide. Dehydration of these hydroxy groups during the thermal treatment process leads to substantial generation of Fe-O-Si bonds, which explains the strongest Fe-SiO_2 interaction in the Fe-Si catalyst. In the case of Fe@Si catalyst, the Fe_2O_3 nanoparticles were first synthesized under the protection of polyvinylpyrrolidone (PVP), thus the Fe_2O_3 nanoparticles were surrounded by a PVP layer, which acted as a barrier between the nanoparticles and SiO_2. Besides, the construction of SiO_2 shell during the Stöber process greatly consumed hydroxy groups, which ultimately lowered the number of hydroxy groups on the SiO_2 surface. These two factors account for the weakest Fe-SiO_2 interaction in Fe@Si catalyst. The content of

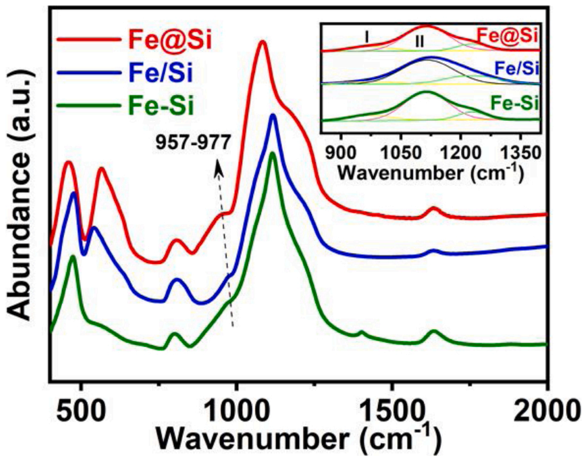

Fig. 3. FT-IR profiles for Fe@Si, Fe/Si and Fe-Si catalysts. Inset pattern is fitting of FT-IR spectra in the wavenumber range 850-1400 cm^{-1}.

hydroxyl groups on SiO_2 surface in Fe/Si catalyst and Fe@Si catalyst is thought to be comparable since SiO_2 is fabricated through the same protocol in these two samples. The difference in the preparation process of the two catalysts mainly lies in the absence of PVP for the fabrication of Fe/Si catalyst, reducing the barrier between Fe_2O_3 nanoparticles and SiO_2. Therefore, the Fe-SiO_2 interaction in Fe/Si catalyst is stronger than that in Fe@Si catalyst.

3.4. Reduction behavior of the fresh catalysts

The reduction behavior of the catalysts is investigated by using H_2-TPR, and the results are shown in Fig. 4. The corresponding H_2 consumptions are listed in Table 2. Generally, two types of reduction processes can be identified in the H_2-TPR profiles of all the catalysts. The first reduction process in the low-temperature region (< 500 °C) are well fitted with two peaks (peaks I and II), and the sum of peak I and II for the Fe@Si, Fe/Si, and Fe-Si catalysts are 0.30, 0.32, and 0.36 mol H_2/mol Fe, respectively. These H_2 consumption value (higher than the theoretical value of Fe_2O_3 to Fe_3O_4: 0.16 mol H_2/ mol Fe; lower than the theoretical value of Fe_2O_3 to FeO: 0.5 mol H_2/ mol Fe) implies that the first reduction process can be assigned to the reduction of Fe_2O_3 to Fe_3O_4 followed by the partial reduction of Fe_3O_4 to FeO [20]. It should be noted that the metastable FeO could be detected at the temperature lower than 570 °C only if the Fe-SiO_2 interaction was present in the catalysts [42]. The FT-IR analysis confirms the presence of Fe-SiO_2 interaction in all three catalysts, which makes partial reduction to FeO in

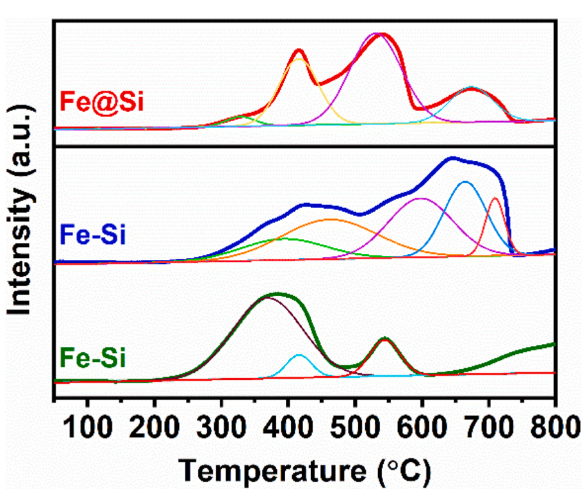

Fig. 4. H_2-TPR profiles for Fe@Si, Fe/Si and Fe-Si catalysts.

Table 2
Quantitative results of H_2 consumption for the catalysts in H_2-TPR.

Catalysts	Peak	Peak Centre [°C]	H_2 consumption [mol H_2/mol Fe] Single	Total
Fe@Si	I	328	0.03	0.95
	II	416	0.27	
	III	531	0.49	
	IV	674	0.16	
Fe/Si	I	340	0.15	1.00
	II	437	0.17	
	III	594	0.32	
	IV	664	0.27	
	V	709	0.09	
Fe-Si	I	369	0.33	0.42
	II	416	0.03	
	III	544	0.06	

the first stage reasonably in the present study. The broad reduction region at high temperature, between 500 and 700 °C, is attributed to the subsequent reduction of FeO/Fe^{2+} species interacting with SiO_2 with different strength to metallic Fe, also including the reduction of remainder Fe_3O_4 [21]. The total H_2 consumption (sum of all the peaks) is usually employed as the indicator of reduction degree that is considered to be governed by Fe-SiO_2 interaction. For the Fe-Si catalyst, there is an obvious reduction peak at high temperature (> 700 °C), indicating the existence of non-reducible species that can hardly be reduced even at harsh conditions (> 1000 °C). Together with the lowest H_2 consumption (0.42 mol H_2/mol Fe), it is safe to further conclude that Fe-SiO_2 interaction in Fe-Si catalyst is strongest as claimed in FT-IR analysis. As shown in Table 2, the H_2 consumption of Fe/Si catalyst (1.00 mol H_2/mol Fe) is higher than that of Fe@Si catalyst (0.95 mol H_2/mol Fe), inconsistent with the Fe-SiO_2 interaction strength order (Fe-SiO_2 interaction strength: Fe@Si catalyst < Fe/Si catalyst). This implies that the reduction behavior, for SiO_2-promoted Fe-based catalyst, is not influenced only by the Fe-SiO_2 interaction. The reduction of iron oxides by H_2 involves two main distinct steps, i.e., the Fe-O bond cleavage and H_2O removal, and the former process was found to be determined by the Fe-SiO_2 interaction strength [21]. Whereas, the efficiency of H_2O removal is related to the pore size, pore structure and morphology of the catalysts [43,44]. For the catalysts obtained from the traditional method (Fe/Si and Fe-Si catalyst), the reduction degree is mainly determined by Fe-SiO_2 interaction strength since the H_2O removal efficiency high (no special configuration between iron oxides and SiO_2). In the case of Fe@Si catalyst where SiO_2 existed as a protecting shell of iron species, the removal of H_2O during the H_2 reduction process is greatly hindered by the SiO_2 shell [44]. Also, the accessible sites for reductant (H_2) can be reduced due to the presence of SiO_2 shell. These two factors obviously degrade the H_2 consumption of Fe@Si catalyst, compared to that of Fe/Si catalyst, even though the Fe-SiO_2 interaction in Fe@Si catalyst is relatively weaker.

3.5. Phase composition of the activated catalysts

Fig. 5a shows the XRD patterns of activated catalysts. As can be seen, Fe_3O_4 and iron carbides dominant the phase structure of all the three catalysts. Both the Fe@Si and Fe/Si catalysts exhibit stronger iron carbides diffraction peaks than the Fe-Si catalyst, indicating that the reduction and carburization degrees of Fe-Si catalyst are suppressed due to the strong Fe-SiO_2 interaction. In order to quantify the specific phase composition of activated catalysts, MES are conducted as shown in Fig. 5b. All the Mössbauer spectra can be fitted with the combination of sextets and doublets. These doublets are assigned to the Fe^{2+} and Fe^{3+} super-paramagnetic (spm) ions on the non-cubic sites with smaller crystallite diameters [45]. The sextets are ascribed to either Fe_3O_4 or iron carbides, and they can be easily identified according to their own

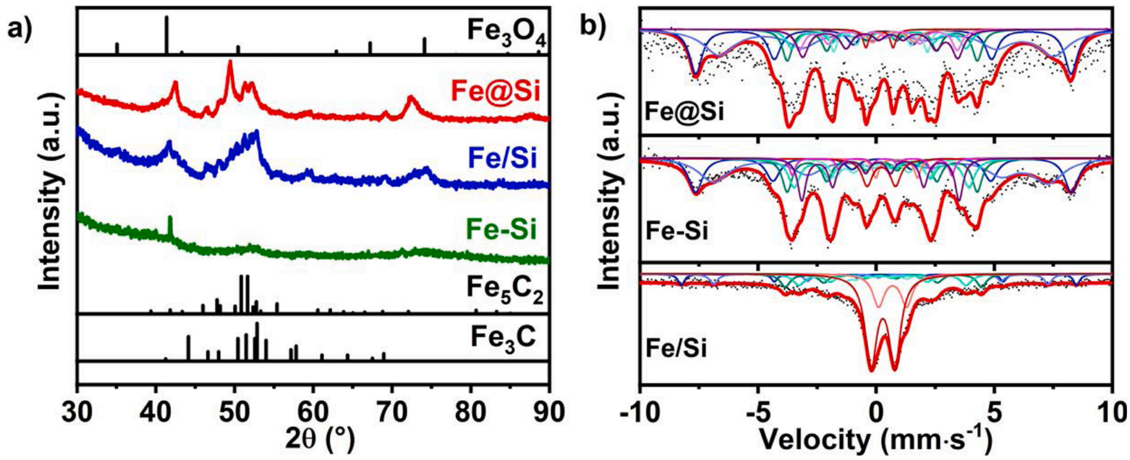

Fig. 5. Fe@Si, Fe/Si and Fe-Si catalysts after activation: (a) XRD pattern and (b) Mössbauer spectra.

hyperfine parameters [20,21]. Based on the Mössbauer parameters summarized in Table 3, it is found that the iron carbides content in Fe@Si catalyst (52.01%) is lower than that in Fe/Si (55.75%) catalyst. As discussed in H_2-TPR section, the SiO_2 shell hindered the removal of H_2O produced during the reduction process, which inhibited the reduction of Fe@Si catalyst regardless of its weaker Fe-SiO_2 interaction compared to Fe/Si catalyst. For the Fe-Si catalyst, the iron carbides content is only 29.67%, much lower than that in Fe@Si and Fe/Si catalyst. This is consistent with the fact that Fe-Si catalyst exhibits the strongest Fe-SiO_2 interaction as confirmed earlier.

3.6. Phase composition of the spent catalysts

Fig. 6a shows the XRD patterns of the catalysts after FTS reaction. As can be seen, the XRD patterns of the three spent catalysts are obviously different from their activated states. Specifically, the intensity of the diffraction peaks assigned to iron carbides in spent Fe@Si and Fe/Si catalysts becomes less evident, while the iron carbides diffraction peaks in the spent Fe-Si catalyst is strengthened. This indicates that the phase transformation behavior during FTS reaction is different for the present three catalysts. Again, MES were conducted to gain the exact phase composition of the spent catalysts (Fig. 6b). According to the fitting results listed in Table 3, the iron carbides content is 41.57%, 49.20%, and 34.78% for the spent Fe@Si, Fe/Si, and Fe-Si catalysts respectively. Compared with the iron carbides content in activated catalyst, it is apparent that iron carbides in both Fe@Si and Fe/Si catalysts are oxidized during FTS reaction, while additional iron carbides formed in spent Fe-Si catalyst. These different phase transitions behavior will be detailed later.

Table 3
Results of Mössbauer fitting for the activation or spent catalysts.

Catalysts	Phase composition					
	FexC[a] [%]	Fe_3O_4[a] [%]	Fe^{2+}/Fe^{3+}[a] [%]	FexC[b] [%]	Fe_3O_4[b] [%]	Fe^{2+}/Fe^{3+}[b] [%]
Fe@Si	52.01	41.18	6.81	41.57	46.27	12.16
Fe/Si	55.75	39.96	4.28	49.20	45.32	5.48
Fe-Si	29.67	3.16	67.17	34.78	3.36	61.86

[a] Activation treatment, 24 h under atmospheric syngas (H_2/CO = 2) at 350 °C.
[b] After reaction for 64 h under 2.0 MPa, GHSV =5000 mL·h^{-1}·g_{cat}^{-1}, syngas (H_2/CO = 2) at 280 °C.

3.7. Morphology of the spent catalysts

Fig. 7 shows the TEM images of Fe@Si, Fe/Si, and Fe-Si catalysts after FTS reaction. As illustrated in Fig. 7a, the spent Fe@Si catalyst almost keeps its original embedded morphology. Moreover, the average size of iron species particles is 45.1 nm, which shows great consistency with fresh Fe@Si catalyst. These results indicate that the sintering of the iron species confined inside the SiO_2 shell is effectively inhibited. However, the images of spent Fe/Si catalyst (Fig. 7b) shows an obvious aggregation of iron species. Also, the iron species in the spent Fe-Si catalyst exhibits slight aggregation (shown in Fig. 7c) despite the strong Fe-SiO_2 interaction exists there. The sintering of catalytic nanoparticles via either Ostwald or particle migration mechanism is, to some extent, inevitable for heterogeneous catalysts, since this process is both thermodynamically and kinetically favored [46,47]. Attempts to alleviate the sintering of nanoparticles includes alloying with the second metal, interaction with supports and confining inside the oxide shells and so on [48–50]. Among them, construction of an oxide like SiO_2 shell outside the active species to form embedded configuration is considered as an efficient strategy to prevent the sintering of active nanoparticles, since their aggregation is physically separated by the oxide shells. For the Fe@Si catalyst in the present study, the presence of SiO_2 shell successfully prevents the iron spices confined from sintering, exhibiting the embedded structure even after the FTS reaction. Whereas, for the Fe/Si and Fe-Si catalysts that prepared through the traditional method, the migration of iron species to form larger particles is inevitable due to the absence of physical barrier originated from SiO_2 shell.

3.8. FTS performance of the catalysts

The FTS reaction for the three catalysts are conducted at the condition of 280 °C, 2.0 MPa, H_2/CO = 2.0, and GHSV =5000 mL•h^{-1}•g_{cat}^{-1}. The CO conversion versus time on stream is presented in Fig. 8a. It is apparent that the three SiO_2-promoted catalysts exhibit varied FTS performance. Specifically, both Fe/Si and Fe@Si catalysts show a higher initial activity (75.0% for the former catalyst and 66.6% for the later one), while Fe-Si catalyst exhibits a lower activity (only 10.1%) at the beginning. With the proceeding of the FTS reaction, the activity of Fe/Si and Fe@Si catalyst declined gradually, however, the activity of Fe-Si catalyst increased steadily to a certain extent (29.7% at 64 h).

Iron carbides are usually identified as the active phase for Fe-based FTS catalysts, and linear correlation of iron carbides content to activity has been frequently reported [2,45]. These catalytic performance discrepancies for the present three catalysts can be related to their varied iron carbides content at different stages. As indicated in the phase composition of the activated catalysts, their iron carbides content after

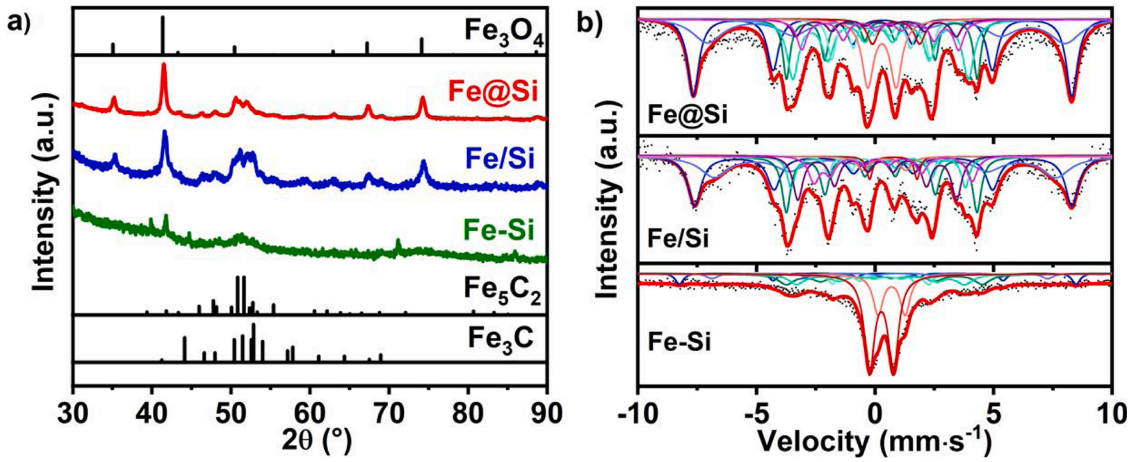

Fig. 6. Fe@Si, Fe/Si and Fe-Si catalysts after reaction: (a) XRD pattern and (b) Moššbauer spectra.

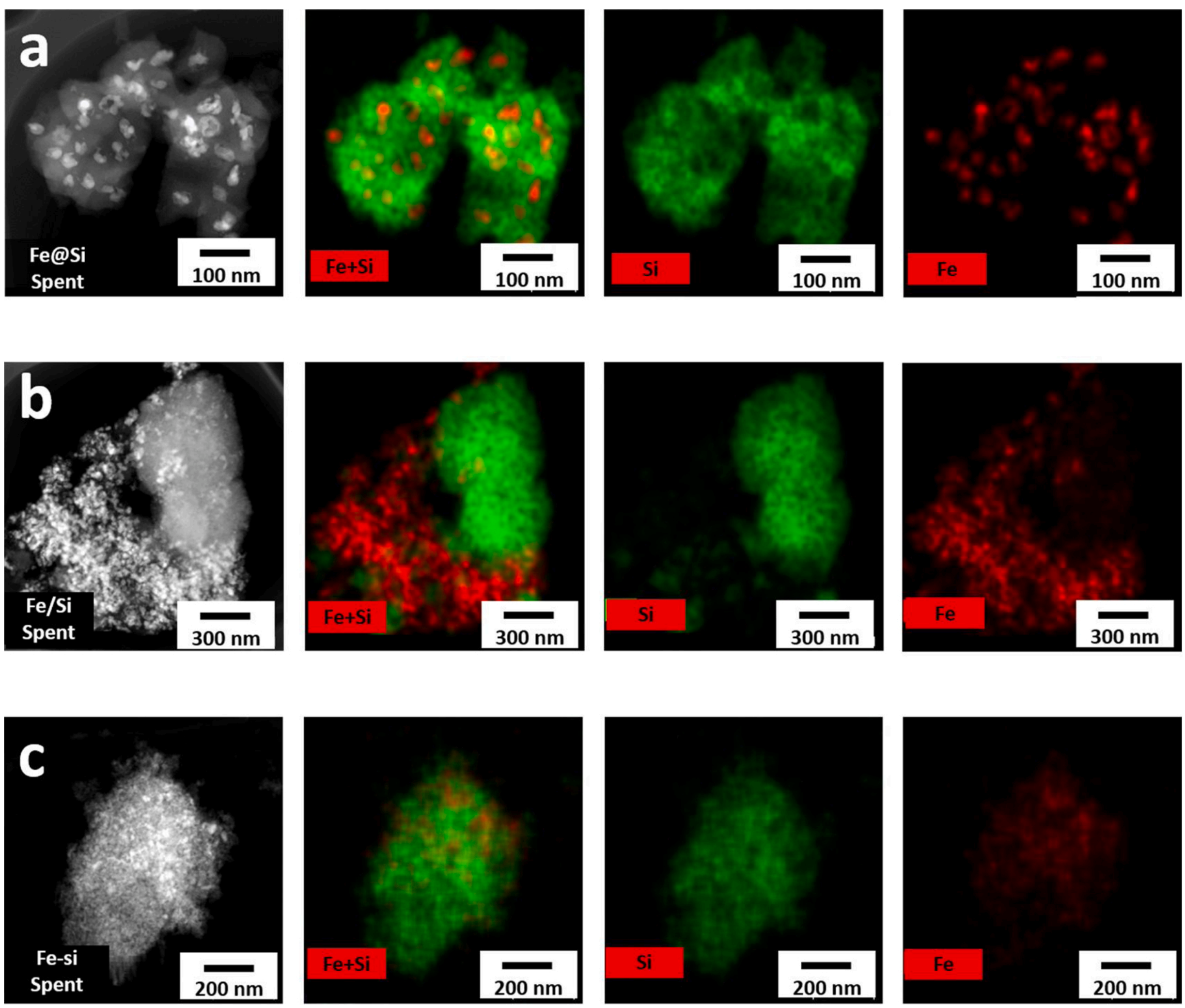

Fig. 7. TEM and high-angle annular dark-field STEM (HAADF-STEM) images of spent catalysts; the elemental maps of the same particle for Si (green) and Fe (red), respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

activation increased with the order of Fe/Si (55.75%) > Fe@Si (52.01%) > Fe-Si (29.67%) catalyst, coincided with the initial activity of the catalysts (Fe/Si > Fe@Si > Fe-Si catalyst). In the course of the FTS reaction, it is reported that additional iron carbides can be produced by further carburization of Fe_3O_4 that has not been carburized in the activation process, while the iron carbides can also be oxidized by H_2O to form Fe_3O_4 simultaneously [51]. These two opposite processes are dynamic and mainly determined by initial iron carbides content, Fe-SiO_2 interaction as well as process conditions [21], thus the phase composition at a certain time is governed by the relative ability of the two

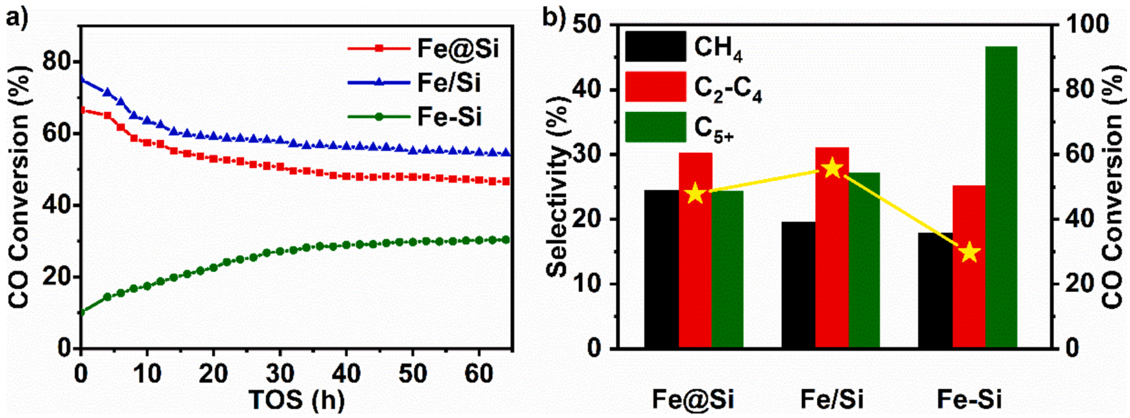

Fig. 8. (a) A stability test of Fe@Si, Fe/Si and Fe-Si catalysts at 280 °C, 2.0 MPa, and GHSV =5000 $mL \cdot h^{-1} \cdot g_{cat}^{-1}$, (b) Comparison of hydrocarbon products distribution for the three catalysts.

processes. For Fe/Si catalyst, the highest initial iron carbides content (55.75%) guarantees its highest CO conversion at the beginning, leading to the higher H_2O partial pressure. Consequently, the oxidation of iron carbides outperforms the further carburization of Fe_3O_4, which accounts for the reduced iron carbides content in spent catalyst (49.20%), compared to that in activated catalyst (55.75%). This partly degrades the activity of Fe/Si catalyst gradually with the FTS reaction going on. Besides, the pronounced iron species sintering of Fe/Si catalyst pays further penalty on its activity. As for the Fe-Si catalyst, the lower iron carbides content (29.67%) after activation leads to its relatively lower initial activity. It is speculated that iron carbides formation through further carburization of Fe_3O_4 dominates the phase transformation during FTS reaction since the iron carbides content increased from 29.67% in activated state to 34.78% in spent Fe-Si catalyst. Therefore, the CO conversion of Fe-Si catalyst increases steadily with the increase of time on stream, despite the slightly iron species aggregation there. In the case of Fe@Si catalyst, similar to that in Fe/Si catalyst, oxidation of iron carbides by H_2O gains advantage over its generation. Moreover, the oxidation process of iron carbides is more prominent according to the difference of iron carbides content in activated and used state. As mentioned earlier, the degraded H_2O removal efficiency caused by SiO_2 shell hindered the reduction process of iron oxides. In the FTS reaction, plenty of H_2O produced compared to the activation process, also, these H_2O cannot be removed promptly due to the confinement effect of SiO_2 shell, increasing the local H_2O partial pressure, which results in the profound oxidation of iron carbides in spent Fe@Si catalyst. Ultimately, the dwindling iron carbides deactivate the Fe@Si catalyst gradually. One may wonder why the confining effect of SiO_2 shell does not exert an obvious positive influence on the catalyst stability, as the iron species stability against sintering has been effectively improved by SiO_2 shell. A possible explanation for this discrepancy is that iron carbides oxidation by H_2O dominates the deactivation factor for Fe@Si catalyst herein.

Fig. 8b illustrates the hydrocarbon distribution versus time on stream of the three SiO_2-contained catalysts, and the detailed product distribution is listed in Table 4. As can be seen, the Fe-Si catalyst exhibits relatively lower CH_4 selectivity and correspondingly higher C_{5+} selectivity, while Fe@Si catalyst displays the highest CH_4 selectivity. The previous study found that, for SiO_2-promoted catalysts, smaller size of iron species is beneficial for the production of heavier hydrocarbons, which is ascribed to the enhanced Fe-C bond that favored the chain growth reaction [20]. This may hold the reason for the highest C_{5+} selectivity for Fe-Si catalyst since the dispersion state of iron species is higher than that in Fe/Si and Fe@Si catalysts. For Fe@Si catalyst, the desorption of hydrocarbons, especially for heavier products, might be hindered due to the confinement effect of SiO_2 shell [43]. These trapped products would experience secondary reaction like deep hydrogenation, thus shifts the products spectrum to lower hydrocarbons for the Fe@Si catalyst. Besides, the presence of SiO_2 shell increased the local H_2O partial pressure as inferred earlier, which would enhance the water-gas shift (WGS) reaction, as confirmed (Fig.S2) by the much higher CO_2 selectivity for Fe@Si catalyst compared to that of Fe/Si catalyst.

Traditionally, the structure-performance relationship is considered to be governed by Fe-SiO_2 interaction for SiO_2-promoted Fe-based FTS catalysts. The present study revealed that the configuration between iron species and SiO_2 exerted a diverse influence on the physico-chemical properties as well as catalytic performance for this kind of catalyst. The construction of SiO_2 shell outside the iron species core inhibits the iron oxides reduction obviously due to the retarding of H_2O removal, despite the weaker Fe-SiO_2 reaction. Also, FTS products desorption is hindered owing to the confinement effect of SiO_2 shell, which influenced the reaction behavior of Fe@Si embedded catalyst profoundly. For example, both the deactivation behavior and the product selectivity variation (higher CO_2 selectivity and lower C_{5+} selectivity) are related to the SiO_2 shell confinement as argued above. Nonetheless, compared to the impregnated and precipitated catalysts (Fe/Si and Fe-Si catalyst), iron species sintering has been effectively inhibited by the physical separation of SiO_2 shell in the catalyst with embedded configuration (Fe@Si catalyst). Further study is suggested to optimize the pore structure of embedded Fe@Si catalyst in order to improve the removal efficiency of either H_2O or hydrocarbons, which would help to alleviate the aggregation of iron particles in traditional SiO_2-promoted Fe-based FTS catalysts, without introducing additional side effects like aggravating oxidation of iron carbides.

Table 4
Performance of Fe@Si, Fe/Si and Fe-Si catalysts after reaction for 64 h[a].

Catalysts	CO Conversion [%]	Selectivity [%]				O/P[b]
		CH_4	C_2-C_4	C_{5+}	CO_2	
Fe@Si	47.9	24.4	30.2	24.3	21.1	1.4
Fe/Si	55.7	19.5	31.0	27.1	22.4	1.5
Fe-Si	29.7	17.9	25.1	46.6	10.5	1.1

[a] Reaction conditions: 280 °C, 2.0 MPa, GHSV =5000 $mL \cdot h^{-1}$·g-1 cat., H_2/CO = 2, the data is collected at TOS =48 h.
[b] The ratio of olefin to paraffin in C_2-C_4 hydrocarbons.

4. Conclusions

In the present study, three distinct SiO_2-promoted iron Fischer-Tropsch synthesis catalysts are prepared to systematically investigate the effect of Fe-SiO_2 interaction as well as morphologies on reduction degree and stability of the catalyst and finally FTS performance. Based on various characterization results, it is found that in spite of the weakest Fe-SiO_2 interaction, the confined iron particles in embedded

(Fe@Si) catalyst exhibits high anti-sintering. However, the construction of the Fe@Si catalyst increases the diffusion resistance of H_2O due to the confinement effect of SiO_2 shell, which strongly inhibited the reduction of iron oxides. Therefore, fabrication of embedded catalyst with large pore sizes and thin SiO_2 shell may be an interesting direction for future research because they can remove H_2O more efficiently to improve reduction degree and finally catalytic performance. On the contrary, the weak Fe-SiO_2 interaction makes the iron species sintered severely in the impregnated Fe/Si catalyst, while the agglomeration of iron is also observed despite the strong Fe-SiO_2 interaction in precipitated Fe-Si catalyst. These results demonstrate the embedded structure plays a crucial role in controlling the strength of Fe-SiO_2 interaction and improving stability of SiO_2-promoted iron Fischer-Tropsch synthesis catalysts. The results of FT synthesis indicate that the Fe@Si and Fe/Si catalysts show an excellent catalytic performance at the very beginning of the FTS reaction, but CO conversion gradually decreased with time on stream. The catalytic activity in Fe-Si catalyst tends to increase. These FTS performances are attributed to variation of iron carbides content at different stages, i.e. the iron carbides are to some extent oxidized by H_2O in Fe@Si and Fe/Si catalysts with diverse morphologies, while the iron species in Fe-Si catalyst are carbonized constantly due to strong Fe-SiO_2 interaction. In summary, this research provides a profound understanding of structure-performance relationships and design strategy for SiO_2-promoted Fe-based FTS catalysts.

Credit authorship contribution statement

Ming Qing, Yong Yang, and Xiao-Dong Wen designed the study. Yu Zhang and Ming Qing performed the most of experiments and data analysis. Hong Wang, Xing-Wu Liu, Suyao Liu, Hongliu Wan, Linge Li, Xiang Gao, and Yong-Wang Li helped revise the manuscript and improve the language. All the authors contributed the idea and participated in the scientific discussions, manuscript comments and corrections.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.cattod.2020.02.026.

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Supporting Information

Comprehensive Understanding of SiO_2-promoted Iron Fischer-Tropsch Synthesis Catalysts: Fe-SiO_2 Interaction and Beyond

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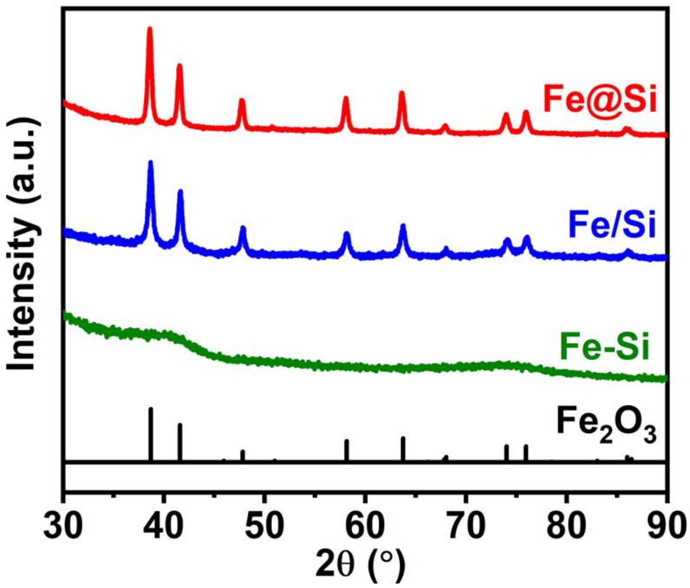

Fig. S1. XRD patterns of fresh catalysts.

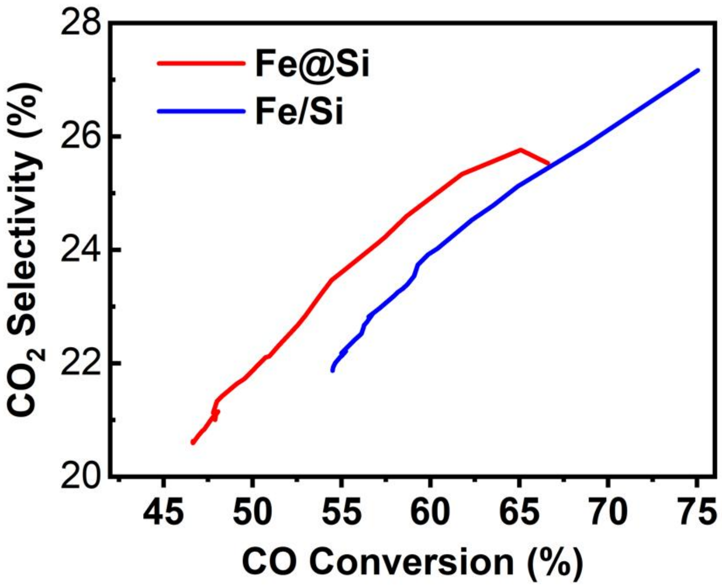

Fig. S2. Relationship between CO_2 selectivity and CO conversion of catalysts.

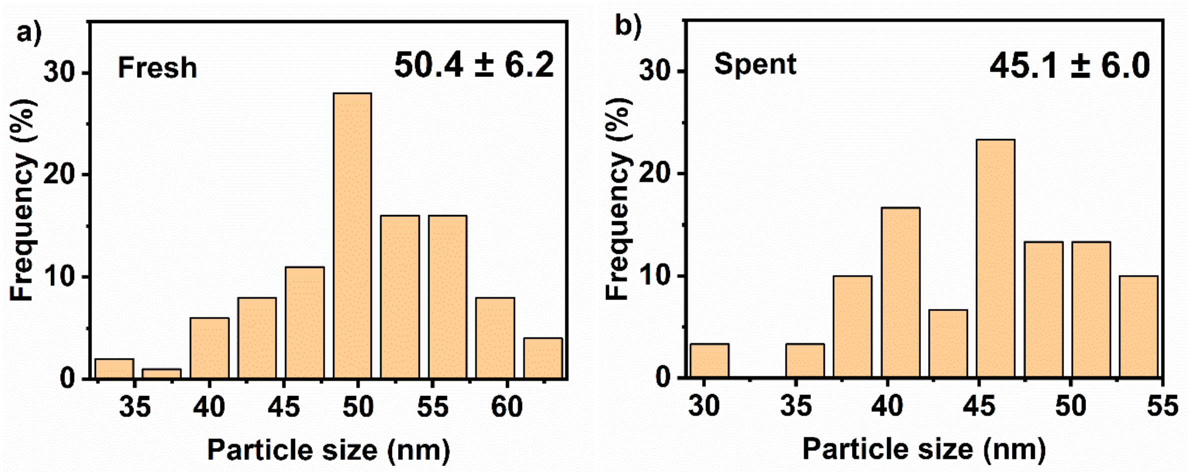

Fig. S3. Iron species particle distributions of fresh and spent Fe@Si catalysts.

Table S1. Quantitative results of Fe-O-Si and Si-O-Si bonds intensity in FT-IR.

Catalysts	Fe-O-Si bond (I)	Si-O-Si bond (II)	Ratio (I/II)
Fe@Si	4.8	62.4	0.077
Fe/Si	8.3	75.8	0.109
Fe-Si	12.3	64.5	0.191